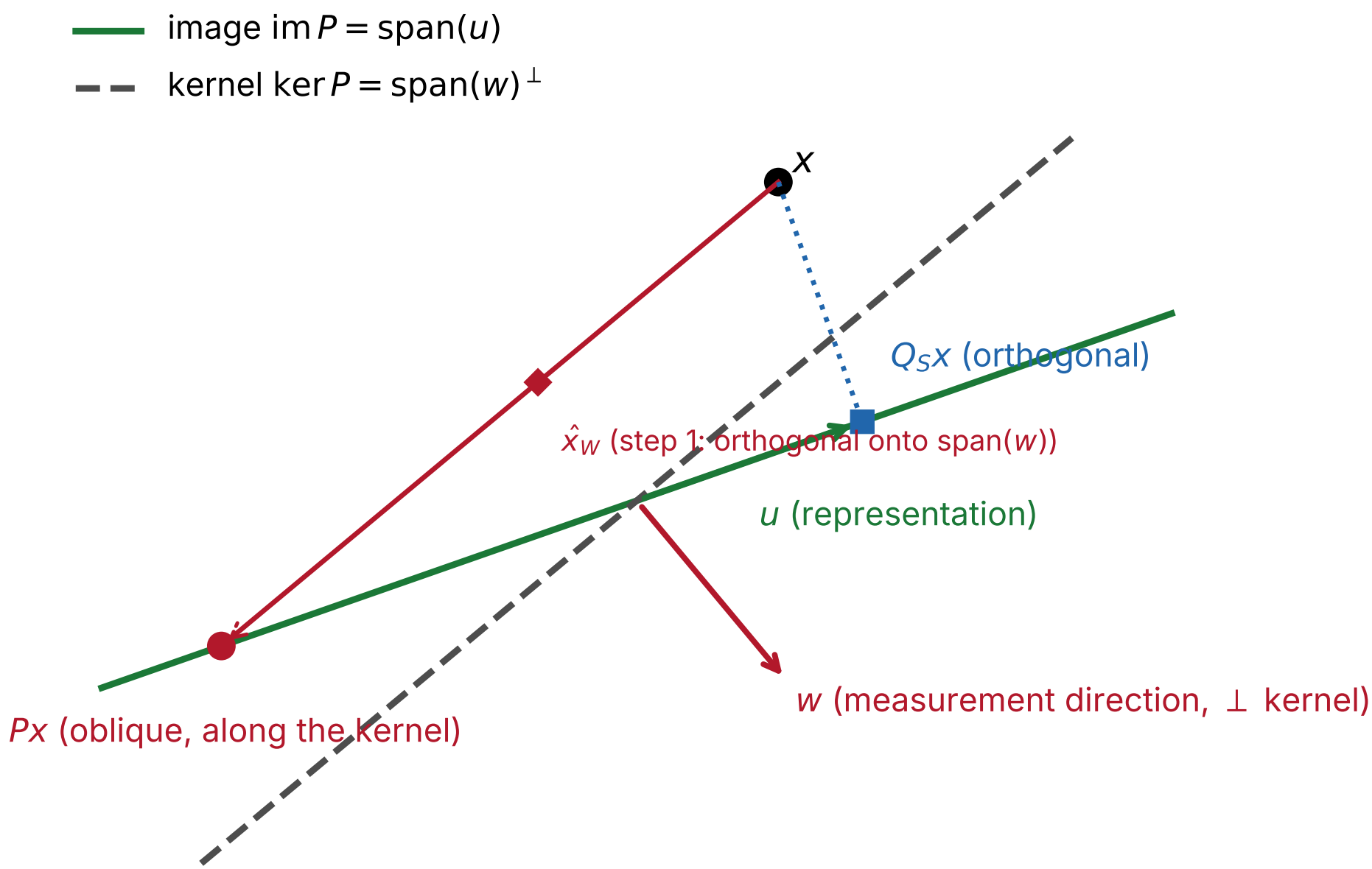
Oblique projection: onto the image, parallel to the kernel



$\theta_{\min} = 21^\circ, \ \|P\|_2 = 1/\sin\theta_{\min} = 1/\cos\angle(u, w) = 2.85$

two steps: project orthogonally onto span(w), then stretch by $1/\cos\angle(u, w)$ along u (step 2)